



Solving Right Triangles Using SOHCAHTOA

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Learning Objectives

- Apply SOH-CAH-TOA to find missing sides of right triangles
- Use inverse trigonometric functions to find missing acute angles
- Evaluate the six trigonometric functions for special angles such as $\pi/3$

Solve each right triangle by finding all missing sides and angles; round side lengths to two decimal places when needed.

1. Find the length of the hypotenuse x of the right triangle with a 30° angle whose adjacent leg measures 12.

Answer: _____

2. Find the length of the side opposite the 45° angle in a right triangle whose hypotenuse is 10.

Answer: _____

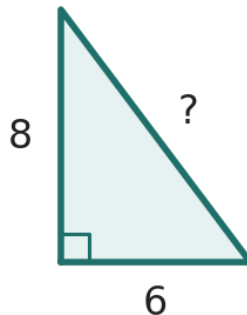
3. A right triangle has a 60° angle with an opposite leg of length 9. Find the length of the adjacent leg.

Answer: _____

4. Find the missing acute angle θ in a right triangle whose opposite side is 7 and hypotenuse is 14.

Answer: _____

5. Find the hypotenuse of a right triangle whose legs measure 6 and 8 using the Pythagorean Theorem, then verify $\sin \theta$ for the angle opposite the leg of length 6.



Answer: _____

6. In a right triangle, the adjacent side to angle θ is 5 and the hypotenuse is 13. Find θ to the nearest degree.

Answer: _____



7. Evaluate the six trigonometric functions of $\theta = \pi/3$ (60°).

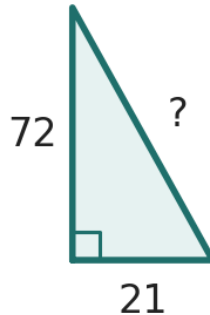
$$\theta = \frac{\pi}{3}$$

Answer: _____

8. A right triangle has angle $A = 25^\circ$ and the side opposite A measures 18. Find the hypotenuse.

Answer: _____

9. For the right triangle with legs 21 and 72, find the hypotenuse and then sin and cos of the angle opposite the leg of length 21.



Answer: _____

10. Find the missing acute angle and the missing side: a right triangle has hypotenuse 20 and one acute angle 35° . Find the side opposite 35° and the remaining acute angle.

Answer: _____





Remind students to identify the opposite, adjacent, and hypotenuse relative to the reference angle before choosing sine, cosine, or tangent.

Solutions

1. Find the length of the hypotenuse x of the right triangle with a 30° angle whose adjacent leg measures 12.

→ Identify the 12 as adjacent to the 30° angle and x as the hypotenuse.

→ Use cosine: $\cos(30^\circ) = \text{adjacent} / \text{hypotenuse} = 12 / x$.

→ Solve for x : $x = 12 / \cos(30^\circ)$.

→ Since $\cos(30^\circ) = \sqrt{3}/2$, $x = 24/\sqrt{3} = 8\sqrt{3} \approx 13.86$.

Answer: $x = \frac{12}{\cos 30^\circ} = 8\sqrt{3} \approx 13.86$

2. Find the length of the side opposite the 45° angle in a right triangle whose hypotenuse is 10.

→ The unknown side is opposite the 45° angle and 10 is the hypotenuse.

→ Use sine: $\sin(45^\circ) = \text{opposite} / \text{hypotenuse} = y / 10$.

→ Solve: $y = 10 \cdot \sin(45^\circ)$.

→ $\sin(45^\circ) = \sqrt{2}/2$, so $y = 5\sqrt{2} \approx 7.07$.

Answer: $y = 10\sin 45^\circ = 5\sqrt{2} \approx 7.07$

3. A right triangle has a 60° angle with an opposite leg of length 9. Find the length of the adjacent leg.

→ Label 9 as opposite and x as adjacent to the 60° angle.

→ Use tangent: $\tan(60^\circ) = \text{opposite} / \text{adjacent} = 9 / x$.

→ Solve: $x = 9 / \tan(60^\circ)$.

→ Since $\tan(60^\circ) = \sqrt{3}$, $x = 9/\sqrt{3} = 3\sqrt{3} \approx 5.20$.

Answer: $x = \frac{9}{\tan 60^\circ} = 3\sqrt{3} \approx 5.20$

4. Find the missing acute angle θ in a right triangle whose opposite side is 7 and hypotenuse is 14.

→ Use sine since opposite and hypotenuse are given: $\sin(\theta) = 7/14$.

→ Simplify: $\sin(\theta) = 1/2$.

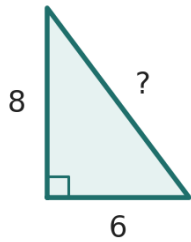
→ Take the inverse sine: $\theta = \sin^{-1}(1/2)$.

→ Therefore $\theta = 30^\circ$.

Answer: $\theta = \sin^{-1}\left(\frac{7}{14}\right) = 30^\circ$



5. Find the hypotenuse of a right triangle whose legs measure 6 and 8 using the Pythagorean Theorem, then verify $\sin \theta$ for the angle opposite the leg of length 6.



→ Apply the Pythagorean Theorem: $c^2 = 6^2 + 8^2 = 36 + 64 = 100$.

→ So $c = \sqrt{100} = 10$.

→ For the angle opposite the leg of 6: $\sin(\theta) = \text{opposite} / \text{hypotenuse} = 6/10 = 3/5$.

Answer: $c = 10$, $\sin \theta = \frac{6}{10} = \frac{3}{5}$

6. In a right triangle, the adjacent side to angle θ is 5 and the hypotenuse is 13. Find θ to the nearest degree.

→ Use cosine: $\cos(\theta) = \text{adjacent} / \text{hypotenuse} = 5/13$.

→ Take the inverse cosine: $\theta = \cos^{-1}(5/13)$.

→ Compute: $\theta \approx 67.38^\circ$, or about 67° .

Answer: $\theta = \cos^{-1}\left(\frac{5}{13}\right) \approx 67.38^\circ$

7. Evaluate the six trigonometric functions of $\theta = \pi/3$ (60°).

$$\theta = \frac{\pi}{3}$$

→ Use the 30-60-90 reference triangle with sides 1, $\sqrt{3}$, 2 (opposite, adjacent, hypotenuse for 60°).

→ $\sin(60^\circ) = \text{opposite}/\text{hypotenuse} = \sqrt{3}/2$; $\cos(60^\circ) = \text{adjacent}/\text{hypotenuse} = 1/2$.

→ $\tan(60^\circ) = \text{opposite}/\text{adjacent} = \sqrt{3}/1 = \sqrt{3}$.

→ Take reciprocals: $\csc(60^\circ) = 2/\sqrt{3} = 2\sqrt{3}/3$, $\sec(60^\circ) = 2$, $\cot(60^\circ) = 1/\sqrt{3} = \sqrt{3}/3$.

Answer: $\sin \theta = \frac{\sqrt{3}}{2}$, $\cos \theta = \frac{1}{2}$, $\tan \theta = \sqrt{3}$, $\csc \theta = \frac{2\sqrt{3}}{3}$, $\sec \theta = 2$, $\cot \theta = \frac{\sqrt{3}}{3}$

8. A right triangle has angle $A = 25^\circ$ and the side opposite A measures 18. Find the hypotenuse.

→ Use sine: $\sin(25^\circ) = \text{opposite} / \text{hypotenuse} = 18 / c$.

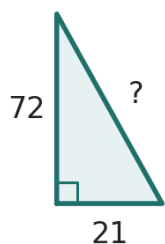
→ Solve: $c = 18 / \sin(25^\circ)$.

→ Compute $\sin(25^\circ) \approx 0.4226$, so $c \approx 18 / 0.4226 \approx 42.59$.

Answer: $c = \frac{18}{\sin 25^\circ} \approx 42.59$



9. For the right triangle with legs 21 and 72, find the hypotenuse and then sin and cos of the angle opposite the leg of length 21.



→ Apply Pythagorean Theorem: $c^2 = 21^2 + 72^2 = 441 + 5184 = 5625$.

→ Take the square root: $c = \sqrt{5625} = 75$.

→ For angle θ opposite the leg of 21: $\sin(\theta) = 21/75 = 7/25$, $\cos(\theta) = 72/75 = 24/25$.

Answer: $c = 75$, $\sin \theta = \frac{21}{75} = \frac{7}{25}$, $\cos \theta = \frac{72}{75} = \frac{24}{25}$

10. Find the missing acute angle and the missing side: a right triangle has hypotenuse 20 and one acute angle 35° . Find the side opposite 35° and the remaining acute angle.

→ The two acute angles must sum to 90° , so the other angle is $90^\circ - 35^\circ = 55^\circ$.

→ Use sine to find the side opposite 35° : $\sin(35^\circ) = a / 20$.

→ Solve: $a = 20 \cdot \sin(35^\circ) \approx 20 \cdot 0.5736 \approx 11.47$.

Answer: $a = 20\sin 35^\circ \approx 11.47$, other angle = 55°

